MEDICAL ROBOTICS PROJECT

# Surgical Workflow Analysis

15-minute slide-by-slide speaker script

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**Christian · slides 1–4 · approximately 3:15**

**Giorgia · slides 5–8 · approximately 3:45**

**Marta · slides 9–11 · approximately 3:10**

**Alessio · slides 12–15 · approximately 4:30**

*Matches the white-theme Canva deck. Slides 16–18 are backup slides and are not part of the timed speech.*

# Christian's section

## Slide 1 — Surgical Workflow Analysis

**Speaker: Christian · Target: 0:20**

Good afternoon. We are Christian Corvino, Giorgia Chiaro, Marta Cudia, and Alessio Olivieri. Our project asks a simple but demanding question: can a model understand where a surgical procedure is in time, not just what is visible in a single frame? We evaluate that question on laparoscopic cholecystectomy videos.

## Slide 2 — Clinical motivation

**Speaker: Christian · Target: 0:45**

A context-aware surgical system must know the current phase before it can provide useful guidance, index a video, or anticipate what comes next. The difficulty is that adjacent phases can look similar in a single frame, while smoke, blood, lighting changes, and tool occlusion make the visual evidence noisy. A short false transition may barely affect overall accuracy, but it can make the inferred workflow clinically implausible. This is why our problem is temporal segmentation, not ordinary image classification.

## Slide 3 — Cholec80 becomes a strict video-level benchmark

**Speaker: Christian · Target: 1:05**

We use Cholec80, which contains 80 laparoscopic cholecystectomy videos with seven phase labels. The original videos are recorded at 25 frames per second. Because phases last minutes and consecutive frames are highly redundant, we sample one frame per second and resize to 224 by 224. This produces 184,578 labeled frames. Crucially, the split is made by whole video: videos 1 to 32 for training, 33 to 40 for validation, and 41 to 80 for testing. No patient or near-duplicate frame can leak across sets.

## Slide 4 — Real surgical duration creates an 11× class imbalance

**Speaker: Christian · Target: 1:05**

The dataset reflects real surgery, so it is strongly imbalanced. Calot triangle dissection and gallbladder dissection together account for more than 72 percent of all frames, while gallbladder retraction is only 3.6 percent. The rarest phase is represented about eleven times less than the most common one. We verified that missing opening or closing phases in a subset of videos are genuine recording characteristics, not parsing errors. Therefore, we preserve the data and manage imbalance during training with class-aware losses rather than artificially changing the test distribution.

# Giorgia's section

## Slide 5 — Every model sees a 2048-D ResNet50 sequence

**Speaker: Giorgia · Target: 0:50**

Every temporal model receives the same kind of input: one 2048-dimensional ResNet50 feature vector for each sampled second. We compare three feature sources. Teammates and original both use frozen ImageNet weights but differ in the extraction implementation. Finetuned adapts the final residual blocks to the laparoscopic phase task before removing the classifier. This common interface isolates the interaction between spatial representation and temporal architecture.

## Slide 6 — Four inductive biases compete on the same sequence

**Speaker: Giorgia · Target: 0:55**

We compare four temporal modeling families. GRU and BiGRU provide compact recurrent baselines, and MS-CBiGRU adds multi-scale convolutions before bidirectional recurrence. The temporal Transformer uses 64-second windows with positional encoding and eight attention heads. MS-TCN applies dilated temporal convolutions to the entire video and iteratively refines predictions. Finally, SF-TMN uses two temporal resolutions with an ASFormer backbone. All produce one phase prediction per second.

## Slide 7 — MS-TCN refines the complete sequence stage by stage

**Speaker: Giorgia · Target: 0:55**

MS-TCN processes the full video in one pass. Within each stage, dilated one-dimensional convolutions expand the receptive field exponentially. The first stage produces an initial sequence; later stages consume the previous probability distribution and refine it. Our multi-task version shares this temporal backbone and adds parallel heads for seven surgical phases and seven instrument labels. We test 10 and 11 dilated layers per stage, both with four stages and 64 channels.

## Slide 8 — SF-TMN reasons at frame and segment scales

**Speaker: Giorgia · Target: 1:05**

SF-TMN explicitly models time at two resolutions. The slow path keeps the original one-second resolution, while the fast path pools groups of 32 frames to reason over segments. The fast representation is upsampled and fused into the slow path with learned weights. In our strongest configuration, both paths use ASFormer blocks with ten layers, 64 channels, local attention, and three decoder refinements. This design combines fine phase boundaries with broader workflow context.

# Marta's section

## Slide 9 — The loss must protect rare phases without creating flicker

**Speaker: Marta · Target: 1:00**

Training must address both class imbalance and temporal fragmentation. For Transformer and MS-TCN we compare weighted cross-entropy, focal loss, and a boundary-aware loss that emphasizes frames near transitions. MS-TCN also uses truncated temporal smoothing, which penalizes abrupt changes in adjacent log-probabilities. The multi-task model adds binary cross-entropy for tool presence with weight 0.40. Recurrent baselines use weighted cross-entropy plus lighter smoothing, while SF-TMN follows its reference setup with unweighted cross-entropy.

## Slide 10 — Every choice is frozen before the test set is opened

**Speaker: Marta · Target: 1:00**

We separate model selection from final evaluation. During development, each feature source, architecture, and loss is trained on the training set. Validation selects the best checkpoint, the loss family, and whether raw output or Viterbi post-processing is better. Only after all these choices are frozen is the 40-video test set instantiated and evaluated once. The instrument threshold is also selected only on validation. This protocol prevents tuning on the test results.

## Slide 11 — Temporal metrics expose errors that accuracy can hide

**Speaker: Marta · Target: 1:10**

Accuracy and macro F1 measure frame classification, but they do not tell us whether the predicted sequence contains too many short, implausible segments. Edit score compares the ordered sequence of segments and strongly penalizes fragmentation. F1 at 10, 25, and 50 percent temporal overlap checks whether predicted phase intervals match real intervals at increasingly strict boundaries. We average those three F1 values as our primary metric. Viterbi decoding is selected only when validation shows that the learned phase-transition model improves temporal quality.

# Alessio's section

## Slide 12 — The accuracy winner is not the temporal winner

**Speaker: Alessio · Target: 1:15**

This plot contains all 21 architecture and feature-source combinations from the final report. The horizontal axis is frame accuracy and the vertical axis is mean segmental F1. A point can move right without moving up, so better frame classification does not guarantee better workflow reconstruction. The 11-layer multi-task MS-TCN with teammates features is furthest right at 92.12 percent accuracy, but its temporal score is 79.65. SF-TMN with original features is highest at 88.36 percent mean segmental F1, with 91.31 percent accuracy. For workflow reconstruction, that is the better trade-off.

## Slide 13 — Full-sequence and multi-scale models dominate temporal quality

**Speaker: Alessio · Target: 1:05**

Looking at the top temporal configurations, SF-TMN with original features ranks first at 88.36 percent. The 10-layer multi-task MS-TCN with finetuned features follows at 86.69 percent, then SF-TMN finetuned at 85.70 and MS-CBiGRU original at 85.67. The temporal Transformer is substantially weaker, between 46 and 51 percent depending on features. Its 64-second windows classify many frames correctly, but they do not reconstruct the global procedure as reliably as full-sequence or multi-scale models.

## Slide 14 — Feature source and depth interact with architecture

**Speaker: Alessio · Target: 1:00**

Feature quality is architecture-dependent. Finetuning helps the Transformer and gives the best macro F1 with the 11-layer MS-TCN, but original ImageNet features are best for SF-TMN and MS-CBiGRU. Depth is also not monotonic. Moving from 10 to 11 layers improves mean segmental F1 by 8.01 points on original features, but reduces it by 2.54 points on finetuned features and 4.21 points on teammates features. These are not pure causal ablations because validation selected different losses. Computationally, all models remain light, with test inference under nine seconds for 40 videos.

## Slide 15 — Conclusion

**Speaker: Alessio · Target: 1:10**

Our answer is task-dependent. If the goal is pure frame accuracy, the best choice is the 11-layer multi-task MS-TCN with teammates features. If the goal is coherent surgical workflow reconstruction, SF-TMN with original features is the stronger model, and that is our recommended configuration. The main limitations are one random seed, one hold-out split, offline future context, and a depth comparison that did not hold every training choice fixed. The next steps are multi-seed evaluation with paired per-video statistics, a controlled 10-versus-12-layer ablation, and a causal model for real-time assistance. The key lesson is that a surgical model is useful only when its sequence is clinically plausible.

# Backup slides

These slides are outside the 15-minute timing. Use them only in response to questions.

## Slide 16 — Frame-level metrics on the test set

**Backup reference**

Backup slide; not included in the 15-minute timing. Use only if the committee asks for the complete frame-level comparison.

## Slide 17 — Temporal segmentation metrics on the test set

**Backup reference**

Backup slide; not included in the 15-minute timing. Use only if the committee asks for the complete temporal segmentation comparison.

## Slide 18 — Auxiliary task, exploratory ablation, and efficiency

**Backup reference**

Backup slide; not included in the 15-minute timing. Contains the remaining quantitative results from Tables 5.3, 5.4, and 5.5.